HATCHSIDE · PRODUCT REQUIREMENTS AND SPECIFICATIONS

HatchSide product requirements and specifications

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	HTS-PRS-001	Version	0.3
Status	Draft	Date	2026-11-05
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/hatchside		
Source	docs/03-requirements.md @ 4004f47		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-11-05	Amish Chadha	Kit 1.9.1 upgrade; R1 to R14 migrated to PR-001 to PR-014 (column Was), each traced to a user requirement in HTS-URS-001; PR-015 to PR-018 added; figures refreshed from the kit 1.9 model
0.2	2026-10-03	Amish Chadha	Populated to TRL 2, then TRL 3 results from the calculation note
0.1	2026-09-30	Amish Chadha	Initial scaffold

HatchSide product requirements and specifications

What the kit must do, each requirement traced to a user need in HTS-URS-001 (links in HTS-RTM-001). Fourteen requirements are the original R1 to R14, migrated to PR-001 to PR-014 with their old numbers in the Was column; PR-015 to PR-018 are new. The requirements describe option A, the tripod and hand winch, which is the current design and is **proposed, awaiting Amish** as the concept (HTS-OPT-001, HTS-DDR-003). On paper 16 of the 18 are met or met by design; PR-008 (grab output) is at risk, at 9.7 L net against 10 L, and PR-011 (corrosion) cannot be shown on paper and waits for a TRL 4 coupon test. None is failed. Option B, the straddle-frame davit, is compared in HTS-OPT-001 and misses PR-005 and PR-006; those statuses are for option A. Method: STANDARDS section 22.

The **reference job** is a 560 mm street manhole 10 m deep, or a household digester opening 2.5 m deep, worked by a crew of two or three with a certified gas detector.

Table 1. Product requirements and specifications. Status from HTS-CAL-001 (kit 1.9 model, 2026-11-05).

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-001	UR-001, UR-004	Safety	No entry needed	Every head shall be fitted, lowered, worked and recovered from outside the opening, with nobody entering	All five heads	pass or fail	n/a	Task walkthrough for every head at a mock manhole, TRL 4; inspection of the model at TRL 3	Met on paper: every head is fitted, worked and recovered from the rim	R1
PR-002	UR-002	Physical architecture	Fits common openings	The tripod shall straddle openings from 0.5 to 1.2 m across	0.5 to 1.2	range	m	Setup trials on sample manholes and hatches, TRL 4; analysis (CAL-001 section 2)	Met on paper: feet on a 3.0 m circle; rim guard inner face 672 mm from the centre	R2
PR-003	UR-003	Performance	Working depth	At least 10 m of rope shall be available below the rim	10	min	m	Winch drum capacity check and depth trial, TRL 4	Met on paper: 15 m of rope; drum capacity to confirm on purchase	R3
PR-004	UR-008	Safety	Working load	The head interface shall carry a safe working load of 150 kg, proof loaded to 1.5 times, with a 1.25 dynamic factor on the proof case	150; 1.5	min; min	kg; factor	Analysis (CAL-001 section 3), TRL 3; proof-load test at TRL 4	Met on paper: leg buckling factor 5.5 at proof with jerk (4.8 with the rope 10 degrees off vertical); rope factor 14.3 at the working load and 4.8 with the winch at stall	R4
PR-005	UR-005	Usability	Fast setup	A crew of two shall set the kit up in 10 minutes or less	10	max	min	Timed setup trial, TRL 4; estimate (CAL-001 section 8)	Met on paper (estimate): 9.0 min	R5

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-006	UR-006	Logistics	Portable loads	No single packed load shall exceed 25 kg	25	max	kg	Weigh each packed load, TRL 4; model masses (CAL-001 section 5)	Met on paper: heaviest load 23.7 kg (grab head); whole kit 153.0 kg in nine loads	R6
PR-007	UR-010	Usability	Quick head change	A head shall be changed in under 2 minutes without tools	2	max	min	Timed changeover trial, TRL 4	Met on paper (estimate): one screw-pin shackle, about 1 min	R7
PR-008	UR-004	Performance	Grab clears silt	The grab shall bring up at least 10 L of wet silt per cycle	10	min	L	Trial in a silt-filled test chamber, TRL 4; analysis (CAL-001 section 8)	**At risk** (estimate): 10.8 L at 70 % fill but 9.7 L after the 10 % that drips back (CAL-001 section 8; STM S9); a bigger grab or a lower target is proposed to Amish (HTS-DEC-001 D16)	R8
PR-009	UR-014	Performance	Depth sample	The sampler shall take a sample whose centre is at a chosen depth within 0.2 m	0.2	max	m	Trial in a layered test tank, TRL 4	Met on paper: line counter reads 0.1 m; tube centre known from the interface	R9
PR-010	UR-003, UR-004	Performance	DomeReach to the floor	The DomeReach tool shall reach 2.5 m	2.5	min	m	Trial at a partner digester, TRL 4; analysis	Met on paper: hinge 2.6 m, tool 3.3	R10

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				below the opening rim (design depth, HTS-DDR-001, D7)				(CAL-001 section 7)	m below the rim	
PR-011	UR-009	Materials	Corrosion resistance	No structural corrosion after 100 hours of salt spray or sewage immersion	100	min	h	Material coupon test, TRL 4	To verify at TRL 4: not verifiable on paper; galvanized steel, 6082 aluminium, A4 fasteners and rope with isolating washers	R11
PR-012	UR-007	Cost	Value-engineering target	Parts shall cost at or near the USD 5,000 value-engineering target (a hypothetical control target)	5,000	max	USD	Priced BOM (`bom/bom.csv`)	Met on paper: USD 2,838, USD 2,162 under the target	R12
PR-013	UR-002	Physical architecture	Heads pass a small manhole	Every head, as it hangs, shall pass a 560 mm opening with 20 mm to spare	540	max	mm	Pass each head through a 560 mm ring, TRL 4; model bounding boxes (CAL-001 section 7)	Met on paper: largest plan diagonal 525 mm (grab); DomeReach passes folded	R13
PR-014	UR-008	Safety	Stable in use	The tripod shall stay stable with the rope 10 degrees off vertical at the safe working load	10	min	degrees	Tilt-pull test during proof loading, TRL 4; analysis (CAL-001 section 6)	Met on paper: tips at 25.0 degrees at the safe working load	R14
PR-015	UR-012	Safety		A rim guard of six hinged	6; 4	min		Inspection of the model	Met by design:	New

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
			Guard the opening	panels with drop pins shall surround the opening, with a splash board on each panel, plus four cones and 10 m of barrier chain for the work zone			panels; cones	and build plan, TRL 3; trial at TRL 4	BOM lines 15, 16 and 22	
PR-016	UR-011	Physical architecture	Made locally	Every made part shall be cut, drilled, bent or welded from stock sheet, plate, bar and tube with a drill and a welder; every other part is bought	No machining	pass or fail	n/a	Inspection of the model, BOM and build plan	Met by design: parts are named with their stock in the D3 model; BOM column make_buy	New
PR-017	UR-013	Usability	Winch effort	One person shall raise and lower a full 150 kg load at a sustained input power no higher than the screening value for hand cranking	75	max	W	Analysis (CAL-001 section 4), TRL 3; crank-force measurement at TRL 4	Met on paper: about 33 W in low gear (31 N at the 250 mm crank)	New
PR-018	UR-008	Safety	Not for lifting people	The winch shall carry a label that it never lifts or lowers a person; the retrieval head shall	Labels on winch and each head	pass or fail	n/a	Inspection of the model labels (BOM line 24)	Met by design: labels on the head and winch; retrieval head is guide-only	New

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				clip a line only, with the haul made on certified rescue equipment					(HTS-DDR-001, D2)	

Categories: physical architecture, performance, materials, usability, packaging and transport, safety, environment and durability, cost. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Assumptions

- Most routine sewer, tank and digester tasks can be done with grab, rake, sampler and pole tools from the rim.
- Crews carry and use a certified gas detector alongside the kit.
- A commodity two-speed hand winch with an automatic load brake is available in the first region.
- Household biogas digesters have an opening of at least 560 mm and a depth of about 2.5 m below the rim; both are to be measured with the first partner.
- The sustained hand-cranking limit of 75 W is the screening value in `.kit/stem_check.py`; an assessment against a cited ergonomics standard belongs to TRL 4.

Safety

> **Safety:** PR-004 and PR-014 are material-handling requirements for tool heads. Nothing in this document qualifies HatchSide for lifting, lowering or rescuing a person, and the winch must never be used for that. A proof load to 1.5 times the safe working load with a competent person present comes before any use over a real opening. Gas testing, traffic guarding and the no-entry rule apply to every trial.